THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — SCIENCE TWO-CHAPTER TEST SET B

Chapters 7 & 8: **Work, Energy and Simple Machines, Sound Waves: Characteristics and Applications** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat, labelled diagrams wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The SI unit of work is: 1

(a) pascal **(b)** watt **(c)** joule **(d)** newton

2. If the velocity of a vehicle doubles, its kinetic energy becomes: 1

(a) four times **(b)** two times **(c)** same **(d)** half

3. The mechanical advantage of a fixed pulley is: 1

(a) 2 **(b)** less than 1 **(c)** 1 **(d)** greater than 2

4. Machines make work easier by changing: 1

(a) the law of conservation **(b)** the energy created

(c) the magnitude or direction of the force **(d)** the total work needed

5. Which of the following is an example of a body possessing only potential energy? 1

(a) a flowing river **(b)** a stretched bow before release

(c) a rolling ball **(d)** a moving car

6. Thunder is heard 5 s after lightning is seen. Taking v = 340 m s^{-1}, the lightning is at a distance of: 1

(a) 3400 m **(b)** 850 m **(c)** 1710 m **(d)** 1700 m

7. In a sound wave, a region of higher density is called a: 1

(a) compression **(b)** rarefaction

(c) trough **(d)** crest of a transverse wave

8. Loudness of a sound depends mainly on its: 1

(a) amplitude **(b)** wavelength **(c)** frequency **(d)** speed

9. The quality of sound that lets us distinguish a sitar from a flute playing the same note is: 1

(a) pitch **(b)** echo **(c)** amplitude **(d)** timbre

10. Ultrasound is used for: 1

(a) making loud noises **(b)** detecting colour

(c) heating food **(d)** imaging internal organs

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 10–15 words each.

11. Define power. Give the SI unit. 2

12. A weightlifter lifts 75 kg through 2 m in 5 s. Find the power (g = 10). 2

13. Name the three ranges of sound frequencies. 2

14. What is an echo? What is the minimum condition to hear one? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 30–40 words each.

15. Differentiate between potential and kinetic energy with an example each. 3

16. State three applications of ultrasound. 3

17. A person raises a 30 cm step with a 40 cm wide ramp. Find the length of the ramp and the mechanical advantage. 3

OR

Why is climbing an inclined plank easier than lifting a load vertically? Does it save work? 3

18. What is timbre? Why do a sitar and a flute sound different at the same pitch and loudness? 3

OR

What is noise pollution? State two harmful effects and two measures. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study:** On a seesaw a child of 20 kg sits 3 m from the fulcrum. 5

(i) Find the moment of the child's weight (g = 10).
(ii) At what distance should a 30 kg child sit?
(iii) Which class of lever is a seesaw?
(iv) What happens if the 30 kg child sits at 3 m?

OR

(i) State the principle of the lever. 5
(ii) Describe the beam balance activity.
(iii) Classify levers with examples.

20. (i) Define timbre and tone. 5
(ii) Why do instruments sound different at the same pitch?
(iii) What is an octave?
(iv) Who studied the acoustics of Indian drums?

OR

Case study: Exposure to loud sound for a long time is harmful. 5
(i) Define noise.
(ii) Give the decibel level of normal conversation and firecrackers.
(iii) State health effects.
(iv) Suggest three ways to reduce noise pollution.

END OF QUESTION PAPER — Total: 40 Marks